Test-sample_textbook - Answer Key

1. What are the two broad categories of cells?

   Answer: B

   Explanation: The textbook states that there are two broad categories of cells: prokaryotic and eukaryotic.

2. What is the main function of the cell membrane?

   Answer: To control what enters and exits the cell

   Explanation: The cell membrane is a selectively permeable barrier that separates the cell's interior from its external environment and controls what enters and exits the cell.

3. The nucleus is the control center of the prokaryotic cell.

   Answer: False

   Explanation: The nucleus is the control center of the eukaryotic cell, not the prokaryotic cell.

4. Define the term 'fluid mosaic model'

   Answer: A structure described as a phospholipid bilayer embedded with proteins

   Explanation: The fluid mosaic model describes the structure of the cell membrane as a phospholipid bilayer embedded with proteins.

5. What is the process by which cells convert nutrients into usable energy?

   Answer: B

   Explanation: Cellular respiration is the process by which cells convert nutrients, primarily glucose, into usable energy in the form of ATP.

6. What are the three main stages of cellular respiration?

   Answer: Glycolysis, the Krebs cycle, and the electron transport chain

   Explanation: Cellular respiration consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain.

7. Glycolysis is an aerobic process.

   Answer: False

   Explanation: Glycolysis is an anaerobic process, meaning it does not require oxygen.

8. Define the term 'chemiosmosis'

   Answer: A process that drives the production of a large majority of the cell's ATP through the electron transport chain

   Explanation: Chemiosmosis is the process by which the electron transport chain drives the production of a large majority of the cell's ATP.